Multiscale Prototype Contrast Network for High-Resolution Aerial Imagery Semantic Segmentation

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***Abstract*— Semantic segmentation of high-resolution aerial images is a challenging task on account of complex scene variations and large-scale differences. However, these two issues are inadequately addressed in general semantic segmentation methods. In this article, we propose a multiscale prototype contrast network (MPCNet) to improve the adaptive capability for different scenes and scales. Specifically, a novel multiscale prototype transformer decoder (MPTD) is designed to extract dynamic scene-specific prototypes as pixel classifiers by fusing information from feature maps and learnable class tokens. To exploit cross-scene context information and accommodate the large-scale difference in the aerial image, we build a multiscale prototype memory queue to store these multiscale prototypes during training. Upon the multiscale prototype memory queue, a novel multiscale prototype contrastive loss is proposed to increase object feature discriminability across multiple scales, which brings better consistency of intermediate features and boosts the convergence of the network. Extensive experimental results on three publicly available datasets demonstrate the effectiveness and efficiency of our MPCNet over other state-of-the-art methods. The code is available at *https://github.com/qixiong-wang/mmsegmentation-mpcnet*.**

***Index Terms*— Aerial imagery, deep learning, prototype, semantic segmentation, transformer.**

I. Introduction

With the development of remote-sensing (RS) technologies, RS images achieve higher spatial resolution with ground object details, which enables pixelwise land cover classification, that is, semantic segmentation. Semantic segmentation for aerial images is a crucial step for automatic scene understanding. It has become a research hotspot with a wide range of applications such as traffic monitoring, urban planning, and road extraction.

In recent years, image semantic segmentation has achieved impressive progress, owing to the development of deep convolutional neural networks (CNNs) [1], [2], [3] and transformers [4], [5], [6]. Typically, fully convolutional networks (FCNs) [3] are the first end-to-end segmentation network, which replaces the fully connection layer in the classification model into 1 × 1 convolution layer to conduct pixelwise segmentation. Consequently, plenty of research has been proposed to improve segmentation performance by increasing the receptive field of neural networks. To explore rich context information, deepLabv3+ [7] employs atrous spatial pyramid pooling (ASPP) consisting of parallel convolutions with different dilated rates, and Chen et al. [8] introduce an adaptive effective receptive convolution (AERFC) for VHR aerial images. To further collect global relational context information, transformer [4], [5] and its optimized variants [9], [10], [11], [12] are developed to capture long-range dependence of object. Recently, it has been reported that transformer-based architecture can achieve superior results for high-resolution aerial imagery semantic segmentation [13], [14]. For example, He et al. [13] embeds the Swin transformer into the classical CNN-based UNet to improve global modeling capabilities. Although the above methods obtain excellent performance by promoting feature discriminative ability, they ignore two specific challenging issues in high-resolution aerial images as follows.

1) The object shows different characteristics (scale, shape, color, etc.) and faces various background interferences due to complex scene variations. As shown in the first row of Fig. 1, the ships in different scenes have enormous intraclass heterogeneity. Typically, the ships on the ground in the left image look more like vehicles than other boats on the shore. The complex scene variation brings an additional challenge to the current segmentation model using a static pixel classifier without cross-scene information for training. Notably, cross-scene information in this article represents the dependencies across different images within the same dataset, which is distinct from the potential meanings of source and target scenes as two different domains.
2) The object has large-scale differences in high-resolution aerial images compared with natural images. As shown in the second row of Fig. 1, the airplanes have tremendous intraclass scale differences due to type and imaging distance variation. This results in a multiscale problem, which makes it difficult to locate and recognize the object on a single scale.

Manuscript received 31 December 2022; revised 25 May 2023; accepted 5 June 2023. Date of publication 6 July 2023; date of current version 24 July 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 41871240 and in part by the National Key Research and Development Program of China under Grant 2022ZD0117400. *(Corresponding author: Jihao Yin.)*

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Digital Object Identifier 10.1109/TGRS.2023.3292919

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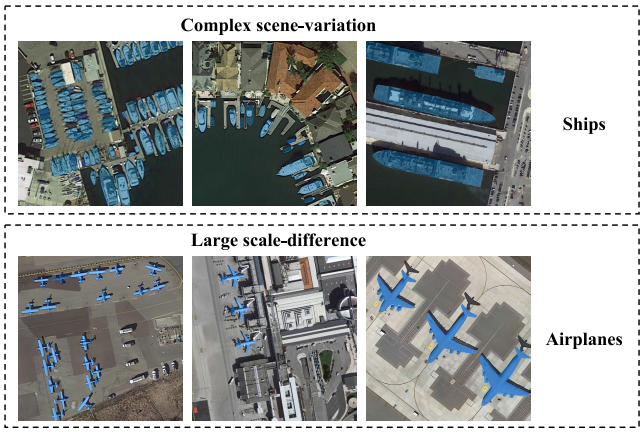


Fig. 1. Illustration of severe challenging issues in high-resolution aerial images. The target objects are highlighted with blue masks. The first and second rows show the ship and airplane objects in different scenes, respectively, demonstrating the complex scene variations and large-scale differences in RS images.

However, the above two issues are inadequately considered in general semantic segmentation methods. On the one hand, most existing semantic segmentation methods employ a 1×1 convolution layer on the last feature map following the paradigm in FCN [3], which only maintains one static prototype for each class across the whole dataset. The learned prototype will be scene-independent during inference so that it is hard to accommodate the complex scene variation in aerial imagery. Moreover, existing methods only explore contextual information within a single scene while ignoring cross-scene contrast. This makes it difficult for the model to adapt to intra-class heterogeneity across different scenes. On the other hand, although several methods propose multiscale feature extraction modules, such as feature pyramid networks (FPNs) [15] and pyramid pooling modules (PPMs) [16], they only regularize the feature space of the last decoder layer. In addition, they neglect the consistency of object feature representation at different scales, which is essential for aerial imagery objects with large-scale variances. As a result, the gradient of the network cannot be directly propagated from the intermediate layer, leading to suboptimal convergence of the network.

In this work, to alleviate the above issues for high-resolution aerial imagery semantic segmentation, we propose a novel multiscale prototype contrast network (MPCNet). It consists of an FPN [15] encoder, a novel multiscale prototype transformer decoder (MPTD), as well as a newly proposed multiscale prototype contrastive loss for training. Specifically, given an input image, MPCNet first employs an FPN encoder to enhance the fusion of cross-scale information and unify the feature channel of different stages. Next, the proposed MPTD extracts multiscale prototypes by fusing information from scene-specific feature maps and learnable class tokens. Notably, the scene-specific prototypes in the last stage of MPTD are also used as the pixel classifier performing on the last feature map, which replaces the traditional static 1×1 convolutional layer for segmentation. This design feature affords the flexibility to employ distinct classifiers tailored to the evolving scene characteristics, thus providing an effective means to mitigate the challenges posed by complex scene variations. To effectively exploit cross-scene context information and improve the multiscale recognition ability of the model, a multiscale prototype memory queue is built to store these prototypes during training. Based on the multiscale prototype memory queue, a novel multiscale prototype contrastive loss is introduced to increase object feature discriminability across multiple scales, which brings better intermediate feature consistency.

To summarize, our main contributions are as follows.

1) We propose a novel MPTD to alleviate the complex scene variation problem in high-resolution aerial imagery, where the pixel classifier is dynamically updated with the input scene by fusing information of multiscale feature maps and learnable class tokens hierarchically.
2) We construct a multiscale prototype memory queue and design a novel multiscale prototype contrastive loss, which introduces multiscale and cross-scene context to accommodate the large-scale differences and scene variations in aerial imagery. It not only brings better consistency of cross-scale features, but also enables the model gradient to be directly propagated from the intermediate layer.
3) We integrate MPTD and multiscale prototype contrastive loss into MPCNet for high-resolution aerial imagery semantic segmentation. A series of experiments are conducted to demonstrate the effectiveness of the proposed MPCNet on three publicly available datasets. Quantitative and qualitative results show that our model outperforms other state-of-the-art methods.

The remainder of this article is structured as follows. Related work is briefly reviewed in Section II. Section III introduces the proposed approach for high-resolution aerial imagery segmentation in detail. The effectiveness of the proposed method and ablation studies are demonstrated in Section IV by experimental results on three real high-resolution aerial imagery datasets. Finally, Section V concludes this article and suggests future research directions.

II. Related Works

A. General Semantic Segmentation

Over the past few years, the breakthrough of deep learning has led to eminent progress in semantic segmentation. FCN [3] is the first end-to-end segmentation framework, which adaptively extracts patterns in the images instead of using traditional hand-crafted features [17]. Subsequently, plenty of variants [5], [7], [18] are proposed to improve the performance. Early improvement focused on preserving detailed spatial information [15], [18], [19], [20], [21]. For example, U-Net [19] concatenates the downsampled feature in the encoder with the upsampled feature in the decoder via designing a U-shaped structure. RefineNet [20] merges multiscale information along the downsampling process to enable high-resolution prediction. Similarly, the FPN [15] merges feature maps of different levels to achieve detailed features with rich contextual information.

Another category of improvement is to increase model contextual information [7], [10], [12], [16], [22], [23], [24], [25].

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To exploit multiscale contextual information, PPM [16] generates multiscale pyramidal feature maps via pyramid pooling, and deeplabv3+ [7] adopts atrous convolutions with different dilated rate to extract features with different receptive field. Different from the multiscale context aggregation model, self-attention [4], [26] and its variants [9], [10], [23], [27], [28] are embedded into the decoder to aggregate global relational context information. Typically, object context network (OCNet) [24] calculates the weighted point-by-point attention map and augments the feature of each pixel by fusing information from all positions. To reduce the computation load and memory usage of the self-attention module, an asymmetric nonlocal neural network [28], attentional class feature network (ACFNet) [10], and object-contextual representation (OCR) [23] explore a group of global descriptors to reconstruct the feature maps instead of treating all pixels themselves as the reconstruction descriptors.

Recently, transformer-based architectures [5], [12], [29], [30] further enhance the long-range context representation ability of feature. Typically, Segmenter [5] jointly processes the patch and class tokens during the decoding phase, which generates a dynamic prototype classifier for pixel segmentation. High-resolution vision transformer (HRViT) [31] enhances ViTs to learn semantically rich and spatially precise multiscale representations by integrating high-resolution multibranch architectures with ViTs. And Mask2Former [30] presents Masked-attention Mask Transformer for panoptic segmentation. Our dynamic prototype extraction transformer (DPET) is inspired by Strudel et al. [5] and Cheng et al. [30], but we adopt lightweight pixel-prototype cross-attention to replace the pixelwise self-attention operation to decrease the computation of matrix multiplication. Compared with [30], our DPET replaces the Masked Attention in [30] with pixel-prototype cross-attention operation. This adjustment allows us to learn class prototypes instead of object instance queries, thereby enhancing learning efficiency for semantic segmentation tasks.

B. Prototype-Based Metric Learning

The concept of the prototype is first used in nearest neighbor classification [32], [33], [34] and deep metric learning [35], [36], [37], [38]. Prototype, also called center [35], or proxy [36], is the one representative feature of a class among training samples. Prototype-based classification methods are straightforward with intuitive interpretation, that is, query observations are directly compared with representative examples. Recently, prototype metric learning has been widely used in few-shot classification [39], [40], [41] and few-shot segmentation [42], [43]. For example, Panet [42] learns class-specific prototype representations from a few support images within an embedding space and then performs segmentation over the query images by matching each pixel to the learned prototypes. Scale-aware detailed matching (SDM) [43] proposes a scale-aware few-shot semantic segmentation network to perform detailed matching with multiple prototypes. Motivated by the above research, we adopt the scene-dependent prototype feature for contrastive learning in high-resolution aerial imagery semantic segmentation, which introduces a global context depiction of the scene.

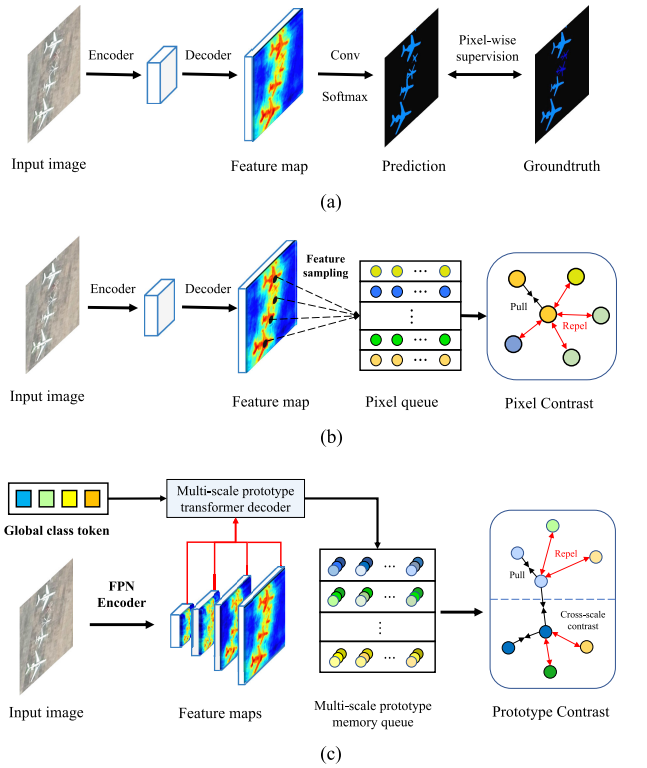


Fig. 2. Illustration of different supervision for semantic segmentation. (a) Pixelwise supervision in single scene and scale. (b) Pixelwise supervision in a single-scale cross-multiple scene. (c) Prototypewise supervision cross-multiple scene and scale.

C. Contrastive Learning

Traditional semantic segmentation methods only perform pixelwise supervision in the last scale on a single scene, as shown in Fig. 2(a). However, they ignore the cross-scene contextual relation, which is vital for high-resolution aerial imagery with complex scene variations. To explore the cross-scene context, recent works have revisited contrastive learning for supervising network feature space by sample pairs contrast [44], [45], [46], [47]. Contrastive learning is first adopted in unsupervised representative learning [44], and it subsequently motivates the joint supervision of contrastive loss and pixelwise cross-entropy loss in semantic segmentation [45], [46]. The process of contrastive learning is shown in Fig. 2(b). Concretely, pixel features of the final decoder layer across different scenes are sampled and stored in a pixel memory queue. Then contrastive loss is conducted on these pixel samples during training to increase feature separability. Although the cross-scene context is introduced by memory queue and contrastive learning, these methods have two deficiencies. First, the feature sampling strategy greatly impacts the segmentation results, as the pixel feature only provides local scene context. Second, distinguishing and depicting objects at multiple scales is crucial for segmentation in many previous research, for example, deep supervision [48], [49]. However, existing contrastive learning for segmentation only constrains the feature in a single scale, that is, the final decoder layer.

To mitigate the above issues brought by sampling pixel features from a single scale randomly, our proposed MPCNet adopts multiscale prototypes as the feature representation for each scene to exploit cross-scene relation information, as shown in Fig. 2(c). This not only eliminates the tricky

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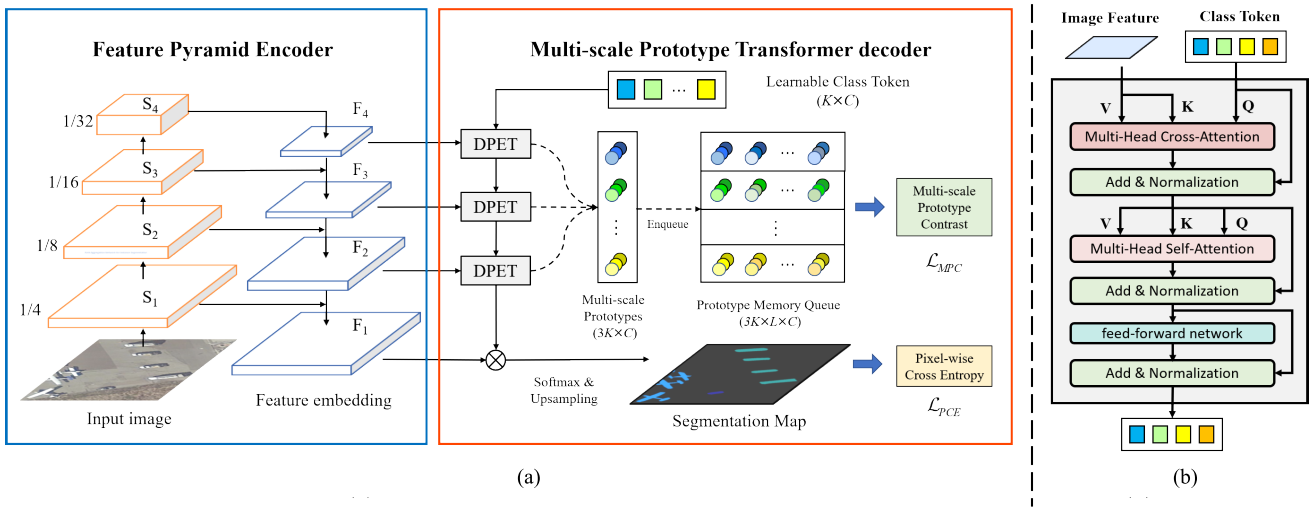


Fig. 3. (a) Flowchart of the MPCNet, which consists of a feature pyramid encoder and MPTD. Given an input image, the encoder generates multiscale feature maps of different resolutions with the same dimension. Subsequently, the decoder adopts three cascade DPETs to extract multiscale scene-specific prototypes and conducts pixel segmentation. To further exploit multiscale context across various scenes, we construct a prototype memory queue during training and propose prototype contrastive learning. (b) Detail structure of the DPET. It consists of multihead cross-attention, multihead self-attention, and feed-forward network operation.

feature sampling strategy, but also introduces multiscale and global context depiction of the scene.

D. Semantic Segmentation of Aerial Imagery

Compared with natural images, aerial images are captured from a bird's view with less common structural information, which induces complex scene variations and large-scale difference challenges. Several works have been proposed to learn more discriminative features for the semantic segmentation of high-resolution aerial images. Foreground-aware relation segmentation (FarSeg) [50] proposes a foreground-aware relation network to address the false alarm issues in high-resolution aerial imagery. To explore long-range dependence within a specific image, relation-augmented FCN (RAFCN) [25] proposes a spatial relation module and channel relation module to learn relationships between any two positions. Hybrid multiple attention network (HMANet) [11] jointly captures global relationships from the perspective of space, channel, and category efficiently. Recently, transformer and hybrid architecture have been introduced to achieve better long-range spatial dependencies modeling. For example, Swin transformer embedding UNet (ST-UNet) [13] embeds the Swin transformer into the classical CNN-based UNet [19] to enhance the feature representation ability of occluded objects. Zhang et al. [14] propose a transformer and a CNN hybrid deep neural network to model the long-range spatial dependence in the image and obtain hierarchical features simultaneously. To improve the segmentation performance for small objects, foreground activation-driven segmentation (FactSeg) [51] proposes a foreground activation-driven framework to address the small object segmentation problem.

However, the above works learn a scene-independent prototype for classification across the entire dataset, which is hard to delineate the intraclass variation for foreground objects in different scenes. Moreover, they only regularize the feature space of the last decoder layer while ignoring the consistency of object feature representation at different scales, which is hard to adapt the scale variance in aerial imagery. In our work, we design MPTD to extract scene-specific prototypes as a pixel classifier, which alleviates the complex scene variation issue. Besides, multiscale prototypes are stored in a memory queue to exploit cross-scale and cross-scene relation information. It not only delivers better consistency of cross-scale features, but also facilitates the gradient propagation of the network.

III. Approach

The overview of the proposed MPCNet framework is shown in Fig. 3. It consists of a feature pyramid encoder and MPTD to conduct pixelwise segmentation. To further exploit multiscale context across various scenes during training, we build a prototype memory queue and propose prototype contrastive learning loss for training.

A. Feature Pyramid Encoder

As shown in Fig. 3(a) (left), given an input image $I \in \mathbb{R}^{H\times W\times 3}$ with its height H and width W, the encoder first generates multiscale feature maps of different resolution $\{S_i \in \mathbb{R}^{2^{i-1}D_i\times(H/2^{i+1})\times(W/2^{i+1})}\}_{i=1}^{4}$ using a backbone network (ResNet or Swin transformer [6]), where D_i is the base channel number of stage i. Rather than directly adopting the backbone feature $\{S_i\}_{i=1}^{4}$ for decoding, we additionally utilize an FPN [15] structure to get feature projection $\{F_i \in \mathbb{R}^{C\times(H/2^{i+1})\times(W/2^{i+1})}\}_{i=1}^{4}$, where $C = 128$ indicates the projected channel number. Formally, FPN operation can be formatted as follows:

$$F^t = \begin{cases} \delta^t\left(S^t\right) + \text{upsample}\left[\theta^t\left(F^{t+1}\right)\right], & t = 1, 2, 3 \\ \delta^t\left(S^t\right), & t = 4 \end{cases} \tag{1}$$

where δ^t and θ^t are transformation functions for dimension reduction, implemented by 3×3 convolution, rectified linear unit (ReLU) [52], batch normalization step-by-step, and upsample denotes bilinear interpolation operation.

Adopting the FPN structure has two benefits. On the one hand, FPN enhances the fusion of high-level contextual information and low-level texture feature, which is vital

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for improving small objects' segmentation performance [53]. On the other hand, it reduces and unifies the feature dimensions in different stages, that is, the output feature map of FPN $\{F_i\}_{i=1}^4$ has the same channel number, which reduces the complexity for the latter dynamic prototype transformer decoder and provides the basis for the multiscale prototype contrastive learning.

B. Multiscale Prototype Transformer Decoder

In the previous scene-independent segmentation framework [3], [12], [22], the segmentation map is directly obtained by applying a 1×1 convolutional layer to the final feature map F_1. However, the classification weights of a 1×1 convolutional layer are fixed after training, which is hard to accommodate the intraclass heterogeneity of objects in aerial images. Inspired by detection transformer (DETR) [54] and its variants [5], [30], we propose a novel MPTD, which extracts scene-specific prototypes for pixel segmentation. As shown in Fig. 3(a) (right), the proposed MPTD takes feature maps $\{F_i\}_{i=1}^4$ generated by feature pyramid encoder as input and outputs multiscale prototype as well as segmentation map. Specifically, given a dataset with N classes, the proposed decoder first extracts multiscale dynamic class prototype vector $\{p_i \in \mathbb{R}^{N\times C}\}_{i=1}^3$ by collecting the information of high-level feature map $\{F_i\}_{i=2}^4$ and learnable class tokens $p_0 \in \mathbb{R}^{N\times C}$ in a hierarchical manner, which can be formulated as

$$p_i = f_{\mathrm{DEPT}}(F_{5-i}, p_{i-1}), \quad i = 1.2, 3 \tag{2}$$

where $f_{\mathrm{DPET}}(\cdot,\cdot)$ indicates the interaction between the feature map and class prototype vector conducted by DPET.

The detailed structure of DPET is shown in Fig. 3(b), which consists of multihead cross-attention, multihead self-attention, and feed-forward network (FFN) operation [54]. Concretely, for the ith stage DPET, given input feature map F_{5-i} and class prototype vector p_{i-1}, the DPET first aggregates the contextual information of input feature and class tokens by multihead cross-attention operation

$$p_i = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{C}}\right)\mathbf{V} + p_{i-1}. \tag{3}$$

Here, $\mathbf{Q} \in \mathbb{R}^N$ is a linear transformation of p_{i-1}, and $\mathbf{V}, \mathbf{K} \in \mathbb{R}^{(HW/2^{12-2i})\times C}$ are the feature transformation of input feature map F_{5-i} achieved by linear projection.

Equation (3) first calculates the similarity matrix $\mathbf{Q}\mathbf{K}^{\mathrm{T}} \in \mathbb{R}^{N\times(HW/2^{12-2i})}$ between each pixel and the class prototype vector and then aggregates a new class prototype vector considering the pixel-prototype similarity. This process achieves the dynamic interaction between the feature map and the class prototype vector. Then the class prototype vectors p_i are further enhanced by multihead self-attention

$$p_i = \mathrm{softmax}\left(\frac{\mathbf{q}\mathbf{k}^{\mathrm{T}}}{\sqrt{C}}\right)\mathbf{v} + p_i \tag{4}$$

where $\mathbf{q}, \mathbf{k}, \mathbf{v} \in \mathbb{R}^{N\times C}$ are linear transformation of p_i. Note that we simplify multihead attention to single-head attention in (3) and (4). And DPET outputs the class prototype vector p_i of the ith stage after the final FFN layer, that is, two layers of 1×1 convolutions with ReLU activation.

After three hierarchical DPET, we obtain the segmentation score masks $\widehat{X} \in \mathbb{R}^{(HW/16)\times N}$ by matrix multiplication of the last prototype vector p_3 and low-level feature map F_1

$$\widehat{X} = \mathrm{softmax}\left(\widehat{F}_1 p_3^T\right). \tag{5}$$

Finally, $\widehat{X}$ is transformed to the final output segmentation map $X \in \mathbb{R}^{H\times W\times N}$ by reshaping and bilinear interpolation. During training, output segmentation map $\widehat{X}$ is constrained to imitate the one-hot ground-truth map $Y \in \mathbb{R}^{H\times W\times N}$ by pixelwise cross-entropy loss $\mathcal{L}_{PCE}$

$$\mathcal{L}_{\mathrm{PCE}} = \sum_{i=1}^{HW}\sum_{n=1}^{N}\mathbb{I}(Y_i == n)\log(X_{ni}) \tag{6}$$

where X_{ni} denotes the output probability of pixel i belonging to class n, and Y_i is the ground-truth label of pixel i, and $\mathbb{I}(\cdot)$ refers to the indicator function as follows:

$$\mathbb{I}(Y_i = n) = \begin{cases} 1, & y_i = n \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

Although pixelwise cross-entropy loss is vital for semantic segmentation tasks, two issues are not well considered. First, cross-scene global context cannot be achieved by single-scene pixelwise optimization. Second, the gradient for intermediate layers is implicitly achieved by back-propagation from the last layer, which induces suboptimal feature discriminability of intermediate layers. To solve the above issues, we further propose multiscale prototype contrastive loss.

C. Multiscale Prototype Contrastive Loss

To explore cross-scene context information and reinforce the mining of hard scenes, we build prototype memory queues $M \in \mathbb{R}^{3N\times L\times C}$ with length L to store class prototype vector of the previous scene, as shown in Fig. 3(a) (right red). The prototype memory queues M are first randomly initialized, and then prototypes in M will be continuously pushed and popped during training following the principle of first-in–first-out (FIFO), that is, once a new scene prototype is pushed to the queue, the earliest scene prototype entering the queue will be popped to maintain the fixed length of the queue. Notably, even if some categories are missing in a specific scene, the MPTD will still extract the class prototype (classifier for segmentation). Thus, the prototype memory queues M will be consistently updated for each class.

Upon the prototype memory queue, we propose multiscale prototype contrastive loss $\mathcal{L}_{MPC}$ to increase feature discriminability by exploiting multiscale context across various scenes. As shown in Fig. 4, $\mathcal{L}_{\mathrm{MPC}}$ consists of cross-scene prototype contrastive loss $\mathcal{L}_{\mathrm{CPC}}$ and scale-aware prototype compaction loss $\mathcal{L}_{\mathrm{SPC}}$. Specifically, for the latest enqueued prototypes (anchor prototypes) $A \in \mathbb{R}^{3N\times C}$, the cross-scene prototype contrastive loss $\mathcal{L}_{\mathrm{CPC}}$ is designed to shorten the intraclass distance and extended the interclass distance

$$\mathcal{L}_{\mathrm{CPC}} = -\frac{1}{3N}\sum_{n=1}^{N}\sum_{s=1}^{3}\log\frac{\exp\left(D_{ap}\left(A_n^l\right)\right)}{\exp\left(D_{ap}\left(A_n^l\right)\right) + \exp\left(D_{an}\left(A_n^l\right)\right)}. \tag{8}$$

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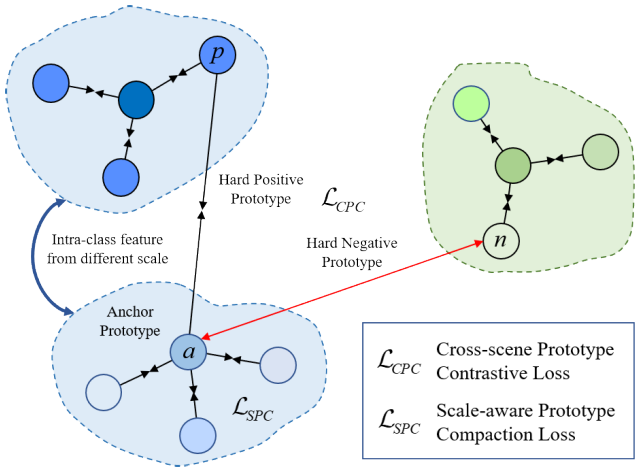


Fig. 4. Illustration of proposed multiscale prototype contrastive loss $\mathcal{L}_{MPC}$, which consists of cross-scene prototype contrastive loss $\mathcal{L}_{CPC}$ and scale-aware prototype compaction loss $\mathcal{L}_{SPC}$.

Here, $D_{ap}(A_k^l)$ and $D_{an}(A_k^l)$ represent the cosine distance of the anchor prototype to the farthest positive prototype and the nearest negative prototype, respectively, which can be formulated as

$$D_{ap}\left(A_n^l\right) = \max_{m \in M_n}\left[1 - \frac{\left(A_n^l\right)^T m}{\|A_n^l\|_2 \|m\|_2}\right] \tag{9}$$

$$D_{an}\left(A_n^l\right) = \min_{\substack{m \in M \\ m \notin M_n}}\left[1 - \frac{\left(A_n^l\right)^T m}{\|A_n^l\|_2 \|m\|_2}\right] \tag{10}$$

where M_n denotes the prototypes in memory queue M with class label n. Notably, different from previous work, which samples pixel feature in the final segmentation stage [45], [46], the proposed cross-scene prototype contrast loss L_{CPC} adopts the multiscale dynamic prototypes for contrastive learning. This not only simplifies the pipeline of pixel feature selection, but also introduces supervision on multiscale feature space. It brings better separability of intermediate features and boosts the convergence of the network.

Equation (8) focuses on increasing interclass separability by jointly optimizing multiple scene prototypes. However, the within-scale intraclass variation is less considered, which is a more common phenomenon in aerial imagery, as shown in Fig. 1. To decrease the variation of intraclass prototypes, we additionally design scale-aware prototype compaction loss L_{SPC} as follows:

$$\mathcal{L}_{SPC} = \sum_{n=1}^{N}\sum_{s=1}^{3}\sum_{l=1}^{L}\left(M_{n,l}^s - \frac{1}{L}\sum_{j=1}^{L} M_{n,j}^s\right)^2 \tag{11}$$

where $M_{n,l}^s \in \mathbb{R}^C$ denotes the lth prototype vector with class n and stage s in memory queues M.

To sum up, the overall objective loss function can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{PCE}} + \underbrace{\lambda_1 \mathcal{L}_{\mathrm{CPC}} + \lambda_2 \mathcal{L}_{\mathrm{SPC}}}_{\mathcal{L}_{\mathrm{MPC}}} \tag{12}$$

where hyperparameters λ_1 and λ_2 are used to balance the loss terms in multiscale prototype contrast loss.

Algorithm 1 Training Process of the Proposed MPCNet

Initialize: Network parameters $\boldsymbol{w}$, prototype memory queue $\boldsymbol{M}$, number of iterations n and learning rate l.
Input: The input image $\boldsymbol{x}$ and corresponding label $\boldsymbol{y}$.
Output: The learned weight parameters $\boldsymbol{w}$ of MPCNet.

for $i = 1$ **to** n **do**
  Encoder:
  Generate basic feature map $\{S_i\}_{i=1}^4$ from the backbone.
  Obtain fused multi-scale feature map $\{F_i\}_{i=1}^4$ from Feature Pyramid Network.
  Decoder:
  Extract multi-scale dynamic class prototype vector $\{p_i\}_{i=1}^3$.
  Generate output segmentation map by matrix multiplication of the last prototype vector p_3 and low-level feature map F_1.
  Update prototype memory queues M using latest prototypes $\{p_i\}_{i=1}^3$ (anchor prototypes).
  Loss:
  Calculate pixel-wise cross entropy loss $\mathcal{L}_{PCE}$.
  Calculate cross-scene prototype contrastive loss $\mathcal{L}_{CPC}$ and scale-aware prototype compaction loss $\mathcal{L}_{SPC}$ following Equation 8 and 11.
  Calculate overall loss $\mathcal{L}$ following Equation 12.
  Update $\boldsymbol{w} \leftarrow \boldsymbol{w} - l\frac{\partial \mathcal{L}(\boldsymbol{w})}{\partial \boldsymbol{w}}$.
end for

To better understand the training process of our proposed MPCNet, we further put an algorithm describing the method in Algorithm 1.

IV. Experiments

To verify the effectiveness of our proposed MPCNet, extensive experiments have been implemented on three publicly available high-resolution aerial imagery datasets, that is, the iSAID dataset [55], International Society for Photogrammetry and RS (ISPRS) Vaihingen and Potsdam datasets.

A. Datasets

1) iSAID: The iSAID [55] dataset has the same image data as the DOTA [56] dataset for object detection. It is the largest aerial instance segmentation dataset with 2806 high-spatial-resolution RS images. Concretely, the training, validation, and test set contains 1411, 458, and 937 images, respectively. The iSAID dataset not only provides instance-level annotation, but also provides semantic mask annotation for segmentation, which is used for our experiment. The images of iSAID are captured from multiple sensors and platforms with multiple resolutions. The original image scale ranges from 800 × 800 to 4000 × 13000 pixels. The iSAID dataset contains 15 categories, that is, ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground field track (GTF), bridge (BR), large vehicle (LV), small vehicle (SV), helicopter (HC), swimming pool (SP), roundabout (RA), soccer ball field (SBF), plane (PL), and harbor (HB). In our

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experiment, we only use the training and validation set for training and evaluating, since the ground truth of the test set is not publicly available.

2) Vaihingen: The ISPRS Vaihingen dataset contains 33 true orthophoto (TOP) tiles and digital surface models (DSMs) with an average size of 2494 × 2064 and spatial resolution of 9 cm, including three spectral bands: near-infrared (NIR), red (R), and green (G) channels. The ground-truth labels involve five foreground object classes (impervious surfaces, buildings, low vegetation, high vegetation, i.e., tree, and cars) and one background class (clutter). Following previous work [11], [14], [57], we use 16 images for training (image IDs: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, 37) and the left 17 images for evaluating. Notably, we only selected the raw TOP images in this work to compare fairly and keep the model general.

3) Potsdam: The ISPRS Potsdam dataset consists of 38 TOP tiles and DSMs with a size of 6000 × 6000 and a spatial resolution of 5 cm. Different from Vaihingen, these TOP files contain four bands: infrared (IR), red (R), green (G), and blue (B). Following [11], [14], and [57], we use 14 tiles with R-G-B channel for testing (image IDs: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13,6_14, 6_15, 7_13) are used for testing and the remaining 24 tiles for training. And the labeling classes are the same as the Vaihingen dataset.

B. Evaluation Metrics

To assess the quantitative performance, three metrics overall accuracy (OA), mean F1 score (mF1), and mean intersection over union (mIOU) are employed in our experiments following [11], [14], [53], [58]:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{13}$$

$$\mathrm{mF1} = \frac{1}{K}\sum_{k=1}^{K}\frac{2\mathrm{TP}_k}{2\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{14}$$

$$\mathrm{mIOU} = \frac{1}{K}\sum_{k=1}^{K}\frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{15}$$

where K is the class number defined in the dataset, and TP, FP, TN, and FN are the number of true positives, false positives, true negatives, and false negatives, respectively.

C. Implementation Details

The implementation is based on PyTorch1.7 and mmsegmentation [59]. All the experiments are conducted on a server with eight NVIDIA GeForce RTX 3090 (24 GB). For the iSAID dataset, we crop the image into 896 × 896 with stride 512 following [50] and [60] for training. As for Vaihingen and Potsdam datasets, the input image is cropped to 512 × 512 with 50% overlap. To avoid over-fitting during training, we adopt a data augmentation pipeline as follows: random resize (ratio from 0.5 to 2), random crop, and random flip (with a probability of 0.5). As for inference, we adopt sliding windows to traverse the whole image to achieve seamless prediction results, and then multiple predictions are averaged to get the final prediction. The sliding window size is set to 896 × 896 for the iSAID dataset with stride 512. As for Vaihingen and Potsdam datasets, we set the sliding window size to 512 × 512 with stride 256.

For training details, we adopt the adaptive moment estimation (AdamW) algorithm as the optimizer with an initial learning rate of 0.00006, a momentum of 0.9, and a weight decay of 0.01. A polynomial learning rate policy is adopted with the formula: $lr = lr_0 \times (1 - i/T)^{0.9}$, where lr_0 represents the initial learning rate, i denotes the current number of iterations, and T denotes the total number of iterations. And the total number of iterations is set to 160k. Additionally, we applied a warm-up strategy in the first 1500 iterations to slow the model's early overfitting of minibatch. And the parameters in the backbone are initialized with pretrained weights to speed up the convergence of the model.

D. Experiments on the iSAID Dataset

1) Comparisons With State of the Art: We report per-class IOU and mIOU on the iSAID benchmark in Table I for a comprehensive comparison. Specifically, 12 typical methods are selected as comparison methods, which can be divided into two parts according to their image encoder. The first part reports the performance of methods equipped with CNNs as the encoder, while the second part presents the methods using Transformer-based structure backbones as the encoder. For the purpose of a fair comparison, we present the results of our MPCNet using both the ResNet-50 and Swin-B backbones. It can be observed that our model surpasses the comparison methods under different backbone settings. Specifically, our model with ResNet-50 backbone surpasses classical Deeplabv3+ [7] and high-resolution network (HRNet) [18] by a large margin. It is mainly because Deeplabv3+ and HRNet only consist of brunches of traditional convolution layers with limited respective fields, which fail to extract global context information. In contrast with the attention-based method HMANet [11], our MPCNet achieves 4.0% improvement in mIOU, which shows the importance of multiscale contrastive learning. Moreover, when the proposed MPCNet is equipped with an advanced Swin-B backbone, our model achieves state-of-the-art performance in all metrics. Compared with the recently proposed Mask2former [30], our model still reaches an improvement of 5.6% in mIOU using the same backbone, which shows the effectiveness of adopting multiscale prototype contrastive loss for alleviating large-scale variation problems. Additionally, our method outperforms the multiscale transformer HRVIT [31] by a margin of 4.1%, which shows that our method alleviates the intraclass heterogeneity problem in aerial images.

2) Ablation for Module Combination: We conduct ablation experiments to investigate the effect of different modules of the proposed network. We first use the vanilla swin-B with a 1 × 1 convolutional layer as the baseline, and the final segmentation result is achieved by directly upsampling the output. The results are reported in Table II, where FPN denotes the feature pyramid network in the encoder, MPTD denotes our proposed multiscale transformer decoder and MPC denotes the proposed multiscale prototype contrastive loss. As shown in Table II, after adopting the FPN structure in the encoder, the performance increases by 1.58% in mIOU and 1.21% in

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TABLE I

EXPERIMENTAL RESULTS ON THE iSAID DATASET. †IN THIS TABLE I INDICATES THAT REPORTED SCORES ARE COME FROM ORIGINAL PAPER

Method	Backbone	IOU per category (%)															mIOU
		SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	(%)
FCN [3]	-	55.8	64.1	33.6	81.8	52.8	18.6	20.4	53.9	41.3	0	32.1	42.6	51.1	72.5	48.8	44.6
Deeplabv3+ [7]	Resnet-50 [2]	59.0	55.2	75.9	84.2	58.5	59.2	32.1	54.5	33.8	31.1	44.2	67.5	73.8	75.7	45.8	59.3
PSPNet [16]	Resnet-50 [2]	65.2	52.1	75.7	85.6	61.1	60.2	32.5	58.0	43.0	10.9	46.8	68.6	71.9	79.5	54.3	60.3
HR-Net [18]	HRNet-W48 [18]	63.9	69.1	67.8	85.9	49.1	43.6	36.0	60.7	36.7	24.5	46.8	33.1	59.6	80.4	53.9	54.7
DANet [22]	Resnet-50 [2]	66.7	74.1	77.9	86.8	52.5	56.1	42.8	63.7	48.6	31.8	46.9	59.4	73.5	83.7	55.8	61.4
OCR [23]	HRNet-W48 [18]	67.6	75.0	78.5	88.5	57.4	59.6	45.3	63.4	49.2	35.7	46.4	66.7	72.3	84.1	55.0	63.0
HMANet† [11]	Resnet-50 [2]	65.4	70.9	74.7	88.7	60.5	54.6	29.0	59.7	50.3	32.6	51.4	62.9	70.2	83.8	51.9	62.6
Farseg† [50]	Resnet-50 [2]	65.4	61.8	77.7	86.4	62.1	56.7	36.7	60.6	46.3	35.8	51.2	71.4	72.5	82.0	53.9	63.7
Pointflow† [60]	Resnet-50 [2]	72.7	74.9	71.2	88.3	63.0	62.3	35.0	64.8	51.8	41.3	52.5	63.3	74.4	85.1	59.5	66.2
MPCNet(Ours)	ResNet-50 [2]	69.9	70.7	78.6	**89.6**	70.0	58.7	40.2	62.6	51.7	41.7	51.3	64.2	74.3	86.2	56.7	66.6
FPN [15]	Swin-B [6]	68.0	70.5	79.6	87.5	59.6	65.2	38.5	62.8	47.2	39.2	51.5	59.9	73.2	84.1	59.5	65.2
Mask2former [30]	Swin-B [6]	71.0	73.4	72.0	87.4	57.4	60.4	28.7	63.7	51.7	39.9	51.1	62.2	75.5	84.2	56.4	64.6
HRViT [31]	HRViT-b2 [31]	72.9	72.0	72.7	88.5	63.4	60.3	30.9	65.2	53.6	41.6	53.4	68.5	72.6	84.9	58.3	66.1
MPCNet(Ours)	Swin-B [6]	**76.5**	**75.6**	**83.4**	89.4	**71.7**	**62.5**	**41.1**	**65.8**	**53.9**	**48.1**	**55.8**	**72.2**	**76.5**	**86.7**	**62.4**	**70.2**

TABLE II

ABLATION OF MODULE COMBINATION ON THE iSAID DATASET

FPN	MPTD	MPC	mIOU(%)	mF1(%)
×	×	×	63.57	76.62
✓	×	×	65.15	77.83
✓	✓	×	67.94	79.86
✓	✓	✓	**70.18**	**81.60**

the mF1 score. This shows the importance of the fusion of high-level contextual information and low-level texture feature in high-resolution aerial imagery. Moreover, by replacing the 1×1 convolutional layer with our proposed MPTD, the mIOU significantly increases from 65.15% to 67.94%. This shows the effectiveness of our dynamic prototype as a pixel classifier, which proves that our method can alleviate the intraclass heterogeneity issue in aerial images. Additionally, after adopting MPC to exploit multiscale and cross-scene information, the mIOU and mF1 score rise to 70.18% and 81.60%, respectively. Notably, by integrating FPN, MPTD, and MPC together, our method significantly outperforms the baseline in both mIOU by 6.61% and the mF1 score by 4.98%, which shows that our modules are complementary to be integrated.

3) Ablation for the Multiscale Prototype Transformer Decoder: To validate the effectiveness of our proposed MPTD module, we conduct experiments to compare the proposed MPTD with three alternative decoders, that is, a traditional 1×1 convolutional layer, a single DEPT, and a mask transformer decoder [5]. The detail of a 1×1 convolutional layer and a mask transformer decoder is shown in Fig. 5. Specifically, the traditional 1×1 convolutional layer adopts a static prototype as a pixel classifier, while the mask transformer decoder employs the transformer to process the patch and class tokens jointly. Notably, our DEPT is similar to the mask transformer decoder, but we replace the pixel-pixel self-attention operation with pixel–prototype cross-attention. This reduces the computational complexity from $\mathcal{O}(C(HW + K)^2)$ to $\mathcal{O}(CHWK)$ compared with the mask transformer decoder [5].

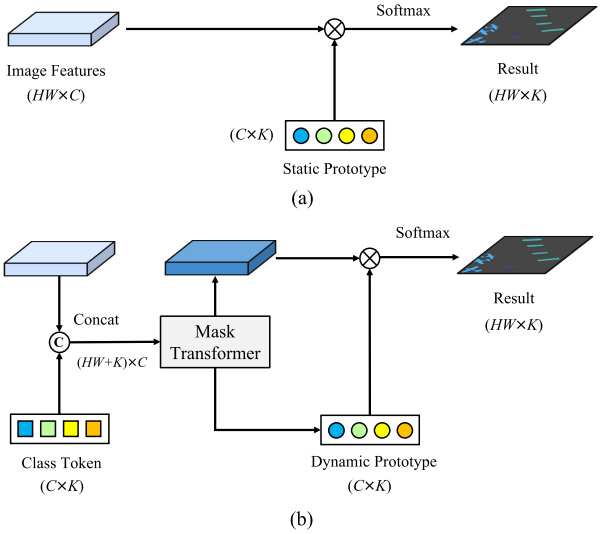


Fig. 5. Illustration of two alternative decoders. (a) Traditional 1×1 convolution layer. (b) Mask transformer decoder.

TABLE III

PERFORMANCE COMPARISON OF DIFFERENT DECODERS ON THE iSAID DATASET

Method	mIOU(%)	mF1(%)	times(ms)
1×1 conv	65.17	77.62	**43.32**
Single DPET	66.26	78.54	44.22
Mask transformer [5]	66.72	78.94	97.13
Hierarchical DPET	**67.94**	**79.86**	46.56

For a fair comparison, we use the same feature pyramid encoder and replace MPTD (Hierarchical DPET) with a different decoder. The results are shown in Table III. Notably, we additionally report inference time on an image with the size of $1 \times 3 \times 512 \times 512$ as efficiency indices. It can be seen that adopting a dynamic prototype classifier brings consistent improvement compared with the traditional 1×1 convolutional layer. Although the performance of a mask transformer is slightly better than adopting a single DEPT, it takes more than twice the inference time. Notably, the performance of

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TABLE IV
ABLATION STUDY OF MEMORY QUEUE LENGTH L

Queue length L	100	200	500	1000	2000
mIOU (%)	69.27	69.62	**70.18**	70.05	69.86
mF1 (%)	80.82	81.13	**81.60**	81.42	81.25

TABLE V
ABLATION OF THE FEATURE SAMPLING METHOD OF MEMORY QUEUE ON THE iSAID DATASET

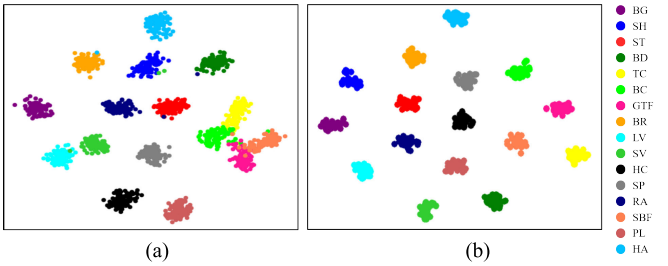
Sampling method	multi-scale	mIOU(%)	mF1(%)
pixel	×	68.62	80.34
pixel	✓	69.11	80.65
prototype	×	69.32	80.73
prototype	✓	**70.18**	**81.60**

MPTD (hierarchical DPET) surpasses the performance of mask transformers with higher efficiency. This shows the effectiveness of multiscale contextual information fusion in high-resolution aerial imagery.

4) Ablation for Multiscale Prototype Memory Queue: As mentioned in Section III-C, we maintain a memory queue $M \in \mathbb{R}^{3N\times L\times C}$ with length L for each class. To analyze the impact of queue length L on segmentation accuracy, we conduct experiments to find the optimal hyperparameter L. The results are shown in Table IV, from which it can be seen that the performance reaches the peak at 70.18% when $L = 500$. However, the performance drops when L is greater than 500. We conjecture storing too many examples might be overfitting to the training dataset.

To evaluate the superiority of our multiscale prototype contrast strategy over existing pixel contrast methods, we compare the performance of different feature sampling methods of memory queue. Specifically, for pixel contrast, pixels of each category are randomly selected from each image in the latest mini-batch following the setting [45]. The results are reported in Table V. It can be seen that using the prototype for contrast outperforms random pixel sampling, boosting the performance from 68.62% to 69.32% in mIOU. This is because the pixel feature only contains the local context of a scene, while the prototype feature provides global depicts of the scene. Moreover, sampling multiscale features for contrast brings consistent improvement in both pixel and prototype contrast and further boosts the performance from 69.32% to 70.18% in mIOU. This shows that depicting and contrasting features at multiple scales is crucial for high-resolution aerial segmentation.

5) Ablation for Multiscale Prototype Contrast Loss: As mentioned in Section III-C, the proposed multiscale prototype contrastive loss consists of two parts of loss at the training stage, that is, cross-scene prototype contrastive loss $\mathcal{L}_{\text{CPC}}$ and scale-aware prototype compaction loss $\mathcal{L}_{\text{SPC}}$, which is controlled by hyperparameters λ_1 and λ_2, respectively. Here, we conduct extension experiments to analyze the influence of weight parameters λ_1 and λ_2 on the segmentation results separately, that is, setting the other parameters to 0. As shown in Tables VI and VII, it can be observed that both $\mathcal{L}_{\text{CPC}}$ and $\mathcal{L}_{\text{SPC}}$ boost the network performance. In general, the model is not sensitive to the change of hyperparameters. The best results are achieved under the setting of $\lambda_1 = 0.8$ and $\lambda_2 = 0.4$. Specifically, simply adding $\mathcal{L}_{\text{CPC}}$ can improve the mIOU from 67.94% to 69.73%, while only introducing $\mathcal{L}_{SPC}$ improves the baseline performance by 69.24%. Based on the above observation, we choose the optimal settings throughout the experiments.



(a) (b)

Fig. 6. Prototype feature visualization of (a) baseline (without MPC) and (b) adding MPC using TSNE [61]. One color stands for one object class's feature.

TABLE VI
ABLATION STUDY OF DIFFERENT HYPERPARAMETER λ_1

λ_1	0	0.6	0.7	0.8	0.9	1
mIOU (%)	67.94	69.35	69.59	**69.73**	69.67	69.48
mF1 (%)	79.86	80.74	81.01	**81.12**	81.06	80.92

TABLE VII
ABLATION STUDY OF DIFFERENT HYPERPARAMETER λ_2

λ_2	0	0.2	0.3	0.4	0.5	0.6
mIOU (%)	67.94	68.71	68.95	**69.24**	69.15	69.03
mF1 (%)	79.86	80.43	80.56	**80.87**	80.73	80.68

To intuitively understand the superiority of multiscale prototype contrastive loss, we conduct feature visualization as shown in Fig. 6. Specifically, we visualize the prototype features extracted by the last DPET in the validation set using t-distributed stochastic neighbor embedding (TSNE) [61]. It can be seen that intraclass prototypes are more consistent and the interclass features are more distinguishable under the supervision of multiscale prototype contrastive loss. Specifically, the prototypes of the tennis court (TC), basketball court (BC), ground field track (GTF), and soccer ball field (SBF) are staggered together in the baseline setting. In contrast, they become separable after adopting multiscale prototype contrastive loss. This shows that our method alleviates intraclass heterogeneity and interclass homogeneity of ground objects in feature space.

6) Qualitative Results: We further visualize the segmentation result in Fig. 7. In the first row, our model can achieve accurate segmentation results of roundabouts in contrast to the comparison methods. In the second row, the comparison methods fail to recognize the soccer ball field in the left top of the image, while our method can precisely distinguish background and foreground, owing to the effective exploitation of cross-scene context. In the last row, the comparison methods mistake some small vehicles for large vehicles. In comparison, our model tackles the intraclass variation problem of small

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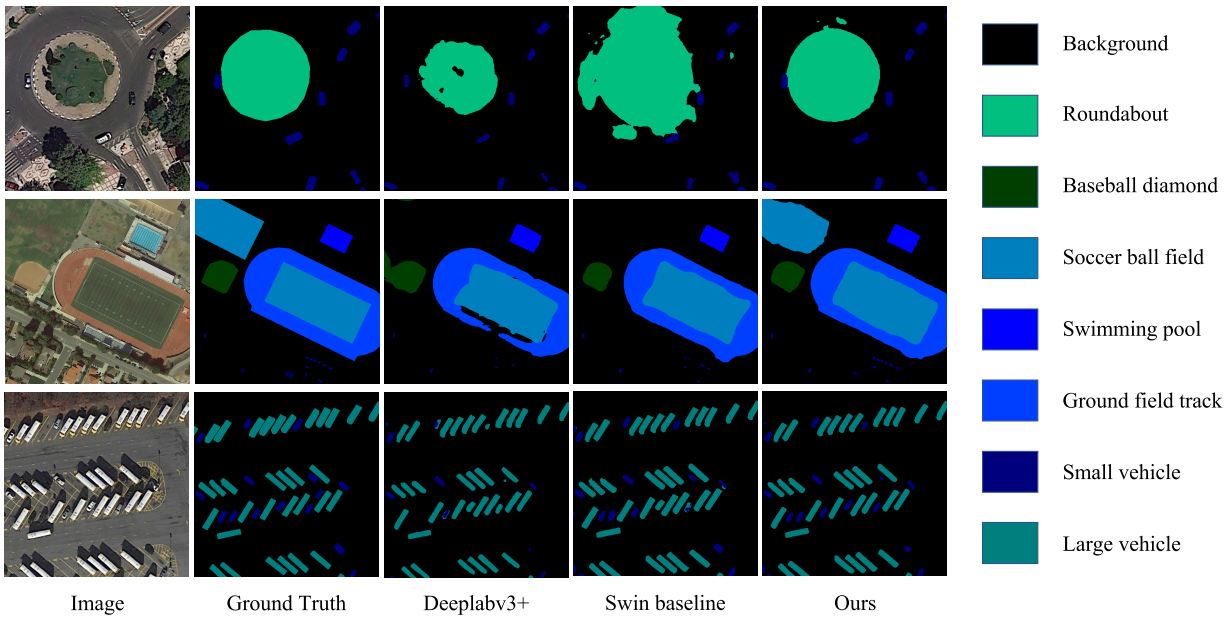


Fig. 7. Qualitative visualizations of segmentation results on image patches of the iSAID dataset.

TABLE VIII
EXPERIMENTAL RESULTS ON THE VAIHINGEN DATASET. †IN THIS TABLE INDICATES THAT REPORTED SCORES ARE COME FROM ORIGINAL PAPER

Method	Backbone	F1 score (%)					mF1 (%)	OA (%)	mIOU (%)
		Imp surf	building	low vegetation	Tree	car			
FCN [3]	-	88.67	92.83	76.32	86.67	74.21	83.75	86.51	72.68
Deeplabv3+ [7]	Resnet-101 [2]	92.38	95.17	84.29	89.52	86.47	89.57	90.56	81.47
PSPNet [16]	Resnet-101 [2]	92.79	95.46	84.51	89.94	88.61	90.26	90.85	82.58
HR-Net [18]	HRNet-W48 [18]	91.70	95.41	83.63	88.52	87.41	89.33	90.52	81.63
DANet [22]	Resnet-101 [2]	91.63	95.02	83.25	88.87	87.16	89.19	90.44	81.32
OCR [23]	HRNet-W48 [18]	92.82	95.41	84.32	90.18	88.26	90.20	90.95	82.72
HMANet† [11]	Resnet-101 [2]	93.50	95.86	85.41	**90.40**	89.63	90.96	91.35	83.49
MPCNet(Ours)	Resnet-101 [2]	93.74	96.14	**85.55**	90.16	89.85	91.09	91.51	83.73
FPN [15]	Swin-B [6]	93.47	96.27	84.82	90.09	88.69	90.67	91.39	83.17
Mask2former [30]	Swin-B [6]	93.23	96.13	85.02	89.93	90.08	90.88	91.32	83.43
HRViT [31]	HRViT-b2 [31]	93.49	96.33	85.31	89.97	90.17	91.05	91.43	83.68
MPCNet(Ours)	Swin-B [6]	**93.65**	**96.49**	85.47	90.28	**90.29**	**91.23**	**91.67**	**84.10**

vehicles. This shows the effect of adopting a dynamic prototype classifier to moderate the problem of intraclass variation.

E. Experiments on the ISPRS Vaihingen Dataset

For the ISPRS dataset, we have selected ten representative methods as comparative references, which comprise seven classical CNN-based methods and three transformer-based methods. The numerical results on the Vaihingen dataset are presented in Table VIII, where the per-class F1 scores are also provided for a detailed performance evaluation across different objects. It is evident that our MPCNet surpasses other methods in all major metrics. Specifically, compared with classical methodologies such as FCN [3] and DeepLabv3 [7], MPCNet exhibits a significant performance improvement. Furthermore, when contrasted with dual attention network (DANet) [22] and HMANet [23] that employ the same ResNet-101 backbone, our MPCNet demonstrates an increase of 1.9% and 0.13% in the mF1 score, and 2.41% and 0.24% in mIOU, respectively. This underscores the superiority of our proposed MPTD over traditional attention-based decoders. Additionally, our model incorporating the Swin transformer backbone surpasses the baseline network FPN, with improvements of 0.56% and 0.93% in mF1 score and mIOU performance, respectively. Importantly, in comparison with state-of-the-art methodologies like Mask2former [30] and HRViT [31], our model delivers an increment of 0.34% and 0.12% in the mF1 score, as well as 0.67% and 0.42% in mIOU. These results lend credence to the efficacy of our proposed multiscale prototype contrastive loss, which leverages cross-scene information for semantic segmentation of high-resolution aerial images.

To achieve an intuitive comparison, samples of segmentation results of Deeplabv3+, Swin transformer baseline, and proposed MPCNet are depicted in Fig. 8. The first row demonstrates that the comparison methods fail to recognize buildings

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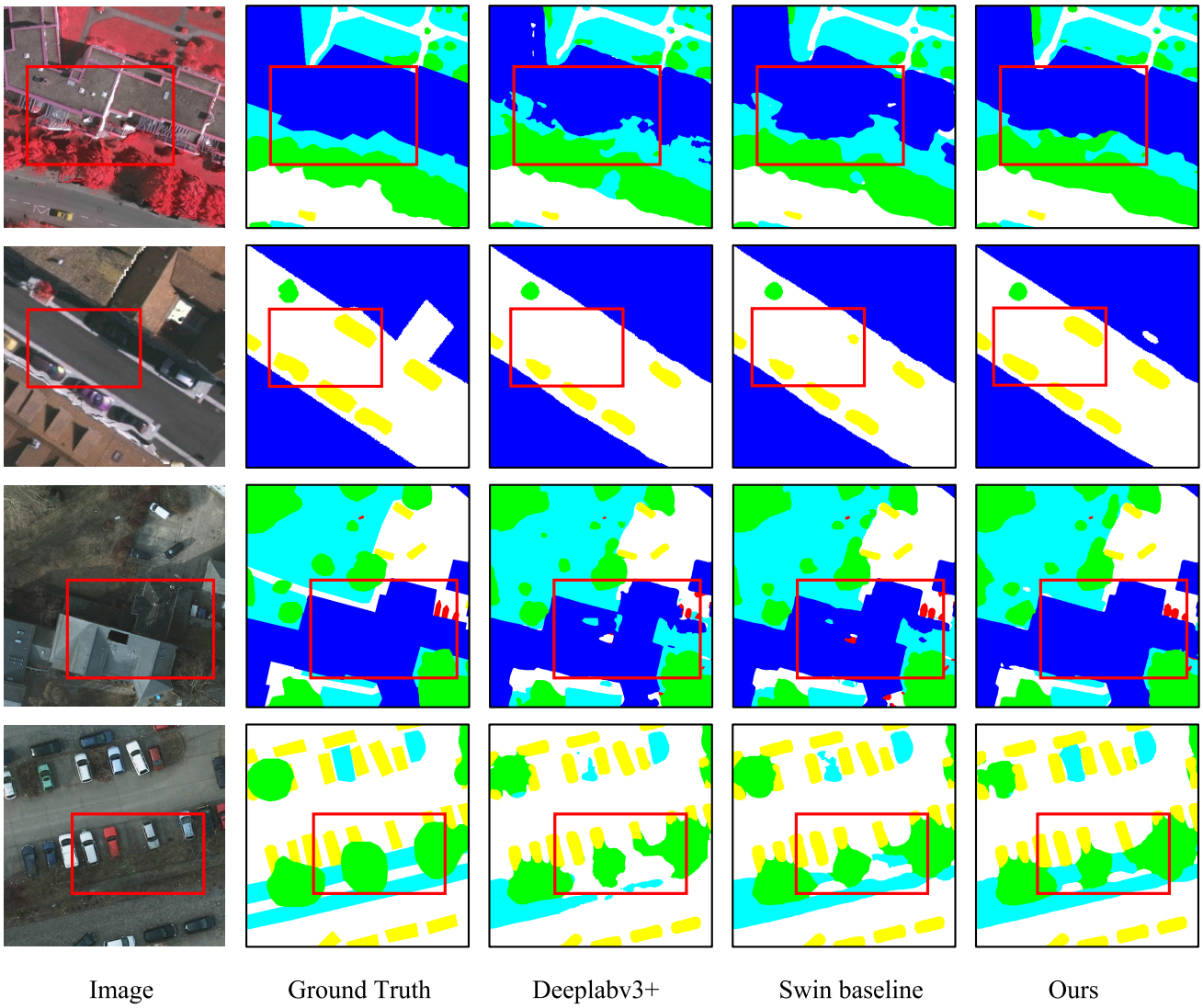


Fig. 8. Qualitative visualizations of segmentation results on image patches of the ISPRS Vaihingen and Potsdam datasets. Classes: Impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and clutter (red).

similar to low vegetation, whereas the proposed MPCNet makes relatively accurate predictions. This indicates that our model has a stronger ability to distinguish ground objects with interclass homogeneity by implementing cross-scene contrast. Additionally, examples in the second row illustrate that MPCNet performs better than other methods when the small vehicle is sheltered by the shadow of the building. This illustrates that our model alleviates the problem of high intraclass heterogeneity in complex scenes by scene-independent pixel classifier. To summarize, through the complex scenes with ambiguous objects, the qualitative comparisons show that our MPCNet provides more accurate labeling than other methods due to the introduction of cross-scene contrast and dynamic pixel classifier.

F. Experiments on the ISPRS Potsdam Dataset

In a bid to further ascertain the efficacy of our network, we carried out additional experiments on the Potsdam dataset. The corresponding numerical results are presented in Table IX. Remarkably, our model incorporating the Swin transformer backbone outperforms other models by a significant margin across various metrics. This outcome attests to the effectiveness of our proposed MPTD and multiscale prototype contrastive loss. Specifically, our MPCNet with ResNet-101 achieves improvements of 0.62% and 0.42% in the mF1 score, and 1.06% and 0.73% in mIOU compared to the competitive attention-based DANet [22] and OCR [23], respectively. When compared with the baseline network FPN employing the Swin transformer backbone, our model delivers improvements of 0.78% and 1.19% in mF1 and mIOU scores, respectively. In contrast to the recently proposed Mask2former [30] and HRViT [31], our MPCNet achieves improvements of 0.73% and 0.56% in mIOU. Notably, MPCNet significantly surpasses other competitors in recognizing the challenging categories of impervious surfaces and trees, which are complicated due to their diverse occurrence and intricate textures under various scenes. This outcome indicates that the application of multiscale prototype contrastive learning can facilitate the learning of more discriminative features, thus leading to better classification accuracy.

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TABLE IX
EXPERIMENTAL RESULTS ON THE POSTDAM DATASET. †IN THIS TABLE INDICATES THAT REPORTED SCORES ARE COME FROM ORIGINAL PAPER

Method	Backbone	F1 score (%)					mF1 (%)	OA (%)	mIOU (%)
		Imp surf	building	low vegetation	Tree	car			
FCN [3]	-	88.61	93.29	83.29	79.83	93.02	87.61	85.59	78.34
Deeplabv3+ [7]	Resnet-101 [2]	92.95	95.88	87.62	88.15	96.02	92.12	90.88	84.32
PSPNet [16]	Resnet-101 [2]	93.36	96.97	87.75	88.50	95.42	92.40	91.08	84.88
HR-Net [18]	HRNet-W48 [18]	93.34	96.35	87.69	89.37	96.32	92.61	91.12	86.82
DANet [22]	Resnet-101 [2]	93.59	96.79	87.60	89.23	96.51	92.74	91.13	86.70
OCR [23]	HRNet-W48 [18]	94.05	97.19	88.08	89.36	96.06	92.94	91.56	87.03
HMANet† [11]	Resnet-101 [2]	93.85	97.56	88.65	89.12	**96.84**	93.20	92.21	87.28
MPCNet(Ours)	Resnet-101 [2]	94.35	97.47	88.72	89.90	96.39	93.36	92.03	87.76
FPN [15]	Swin-B [6]	93.54	97.21	88.21	89.24	96.41	92.92	91.63	87.16
Mask2former [30]	Swin-B [6]	94.32	97.23	88.32	90.12	96.46	93.29	91.83	87.62
HRViT [31]	HRViT-b2 [31]	94.41	97.41	88.76	90.03	96.35	93.39	92.10	87.79
MPCNet(Ours)	Swin-B [6]	**94.54**	**97.63**	**88.99**	**90.44**	96.79	**93.70**	**92.32**	**88.35**

Furthermore, visual results are presented in Fig. 8. As shown in the third row, the aerial scene is complicated, where the buildings, trees, clutter, and roads show interclass similarity. This brings an enormous challenge for the comparison methods, while our model can distinguish similar objects of different semantic categories. In the last row, albeit the vehicle is surrounded by intricate vegetation and tree, the boundaries between areas of trees and car in the segmentation result tend to be more precise and smoother for the case of MPCNet, which shows the effectiveness of our multiscale prototype contrastive loss.

V. CONCLUSION

This article proposes a novel MPCNet for high-resolution aerial image semantic segmentation to improve model adaptive capability across different scenes and scales. In this framework, scene-specific prototypes are extracted as pixel classifiers by fusing information from feature maps and learnable class tokens. Upon the scene-specific prototypes, we construct multiscale prototype memory queues and design a novel multiscale prototype contrastive loss, which introduces multiscale and global context across different scenes. The ablation studies and comparative experiment results verify the superiorities of our proposed framework in terms of high labeling accuracy and computation efficiency. On three publicly available datasets, both quantitative and qualitative results show that our MPCNet provides more precise labeling results than other compared well-known algorithms.

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